

	Work Instruction	Doc. Revision	01
	AXI V810 X-Axis Teknic Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	11 Dec 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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X-Axis Teknic Servo Motor Removal

1. Unload if any board inside machine.
2. Power off the SSCA main power # Importance.
3. Open all front access door for ease of removal process.
4. Manually position the stage to desire position for ease of removal process.
5. Disconnect connector from New X axis Power & Rotary encoder (P/N, 4XASC-A124) as shown.

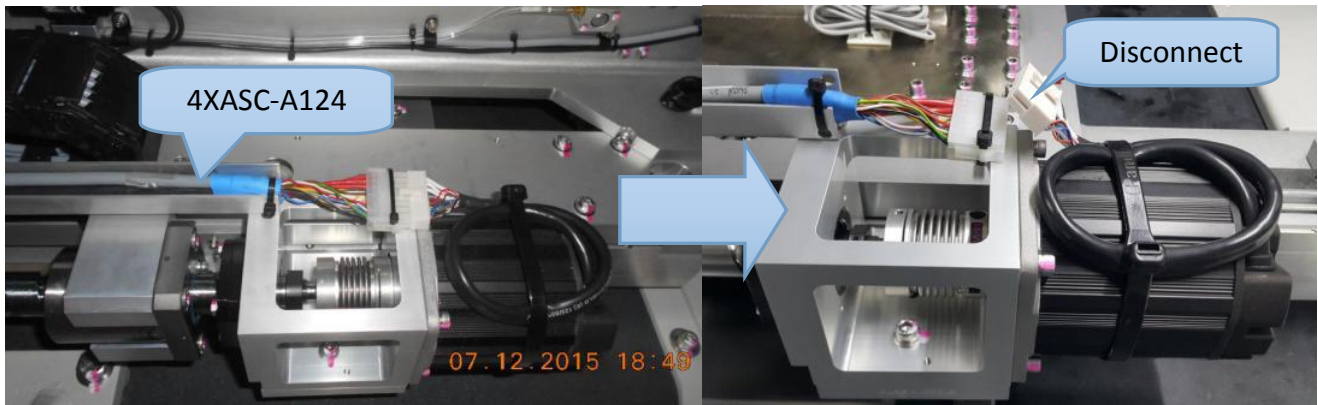


Figure 1 Connector removal

6. Loosen 1 x screws at Bellow Coupling_Bore Dia: 12.7mm/12.7mm (P/N, 2660-0032) as shown.

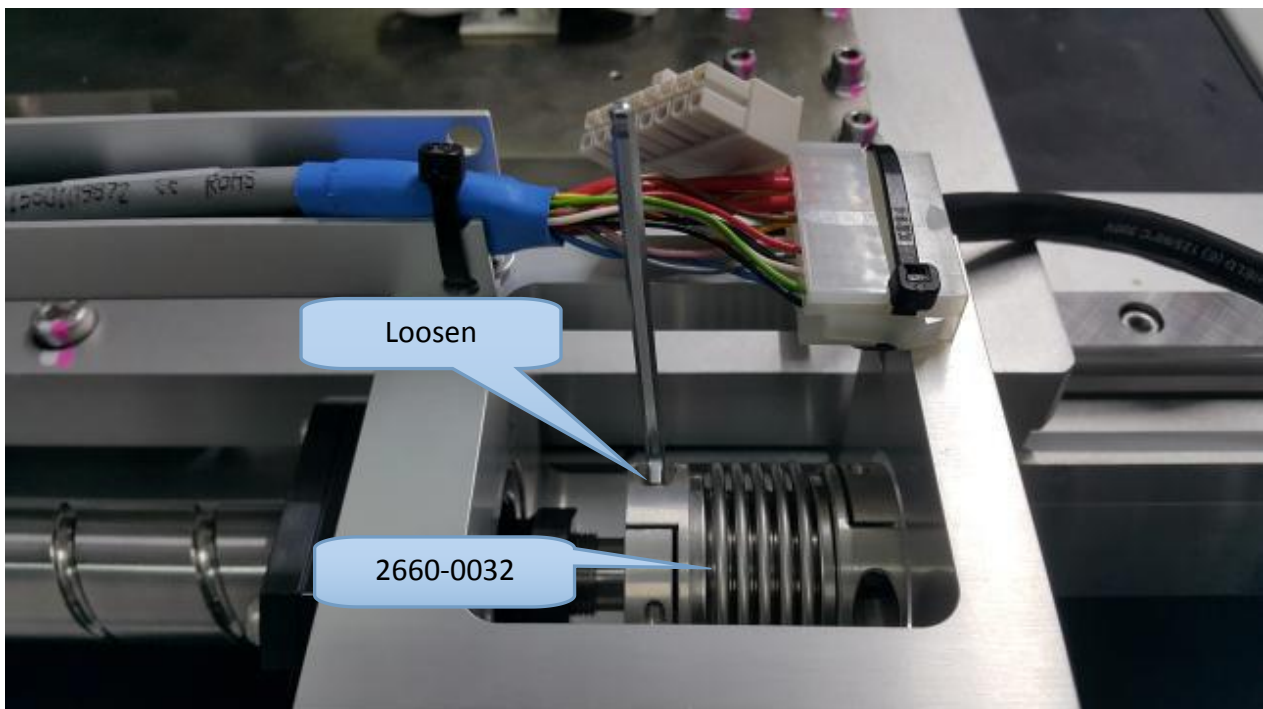


Figure 2 Coupling screws

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7. Unscrew 4 screws at HOUSING, MOTOR X-AXIS (P/N, 4XAXY-010) as shown. Caution, the inner screw for lower part for housing motor need a longer Allen wrench (M5) to remove.

CAUTION, DO NOT worn out Socket head cap screw while unscrew.



Figure 3 Servo Motor screws loosen

8. Gently remove servo motor with coupling from 4XAXY-010 (Motor Housing)

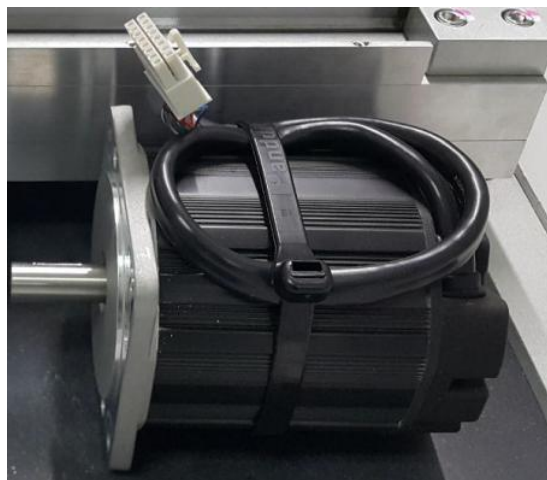


Figure 4 Servo Motor removal

9. Loosen 1 x screw at other end of 2660-0032(Coupling) and gently remove as below Figure 5

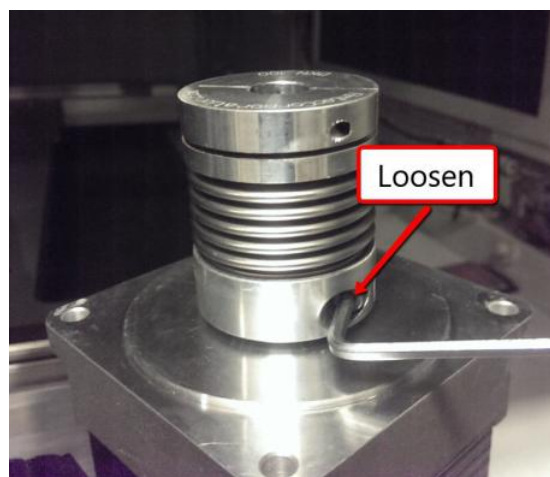


Figure 5 Coupling removal

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X-Axis Teknic Servo Motor Replacement

10. Prepare new NEMA 34, Brushless Rotary Servomotor (Teknic Motor) (P/N, 2670-0035) as shown.

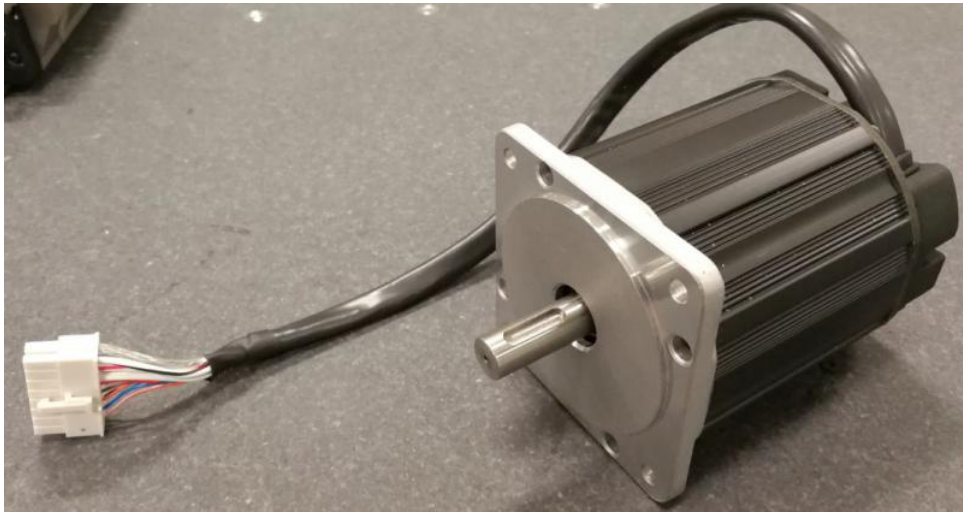


Figure 6 Teknic Servo Motor

11. Assemble 2660-0032 (Coupling) to V810 X-Axis Ball screw Kit (P/N, 4XASP-A024) and tighten 1 x screw only at coupling (End shaft of X axis Ball Screw).

Special remarks :

Both screws at coupling require to be visible

Socket head cap screw could not worn out

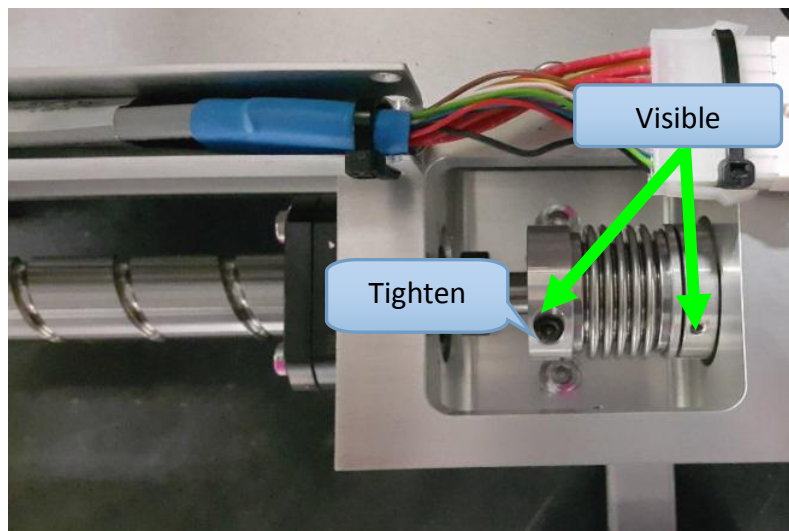


Figure 7 Coupling fastening

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12. Assembly the 2670-0035 (Teknic Motor) to 4XAXY-010 (Motor Housing) as shown & tighten up 4 x screws.

Special remarks :

Socket head cap screw could not worn out



Figure 8 Servo motor assembly

13. Tighten screw at the other end of 2660-0032 (Coupling) as shown.

Special remarks :

Socket head cap screw could not worn out

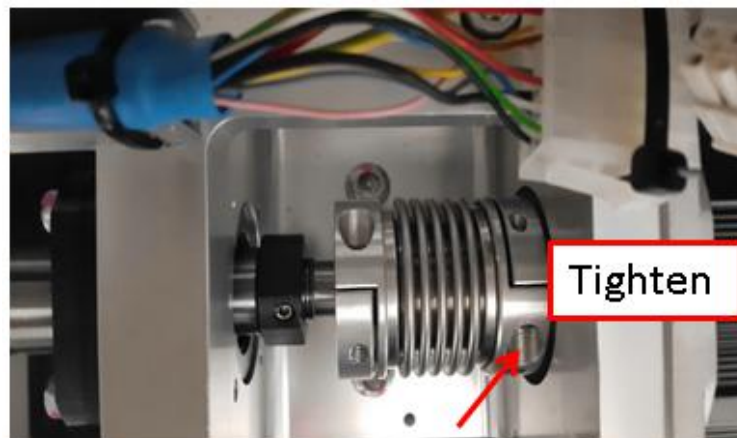


Figure 9 Coupling screw fastening

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14. Route the cable as shown and apply cable to tidy up the extra length cable.

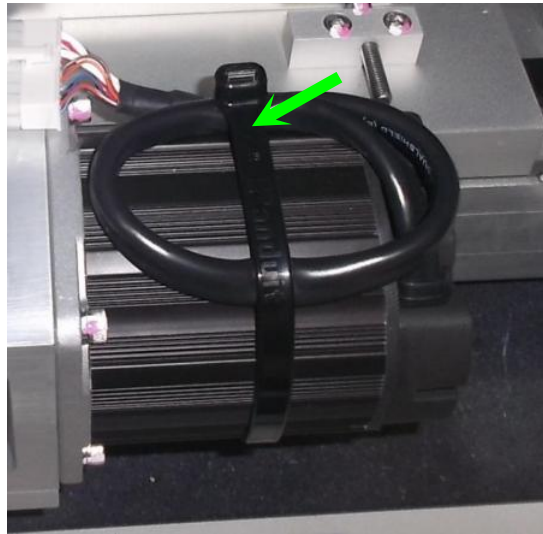


Figure 10 Motor Cable Routing

15. Connect back the motor connector as below Figure 11.

Remarks :Motor connector require properly tie up with cable tie



Figure 11 Motor connector assemble

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16. Remove all tools from machine stage.
17. Ensure all screws are fully tightened.
18. Manually exercise X axis stage in “Left to Right “or “Right to Left” for few routes. . If alignment is require, may refer to procedure **“AXI V810 X-Axis Ball Screw Alignment Procedure”**
19. Close all the access door.
20. Go to Panel Positional, load Long scan.path and run 10 loops looping test.
21. Run 1 loop of CD & A.